

FIE401: Financial Econometrics

Course Introduction

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What is a goal of this lecture?

To help you to make an informed decision on course participation.

Course in a nutshell

- ▶ Course content:

$$y = \beta_0 + \beta_1 x + u$$

- ▶ Course schedule (12 weeks)
 - ▶ Video lectures
 - ▶ Coding exercise
 - ▶ 3 hours class with 1h Q&A and 2h lab
- ▶ Course approval:
 - ▶ 3 assignments (Pass/Fail)
- ▶ Exam:
 - ▶ Written report (60%)
 - ▶ Oral exam (40%)

Overview

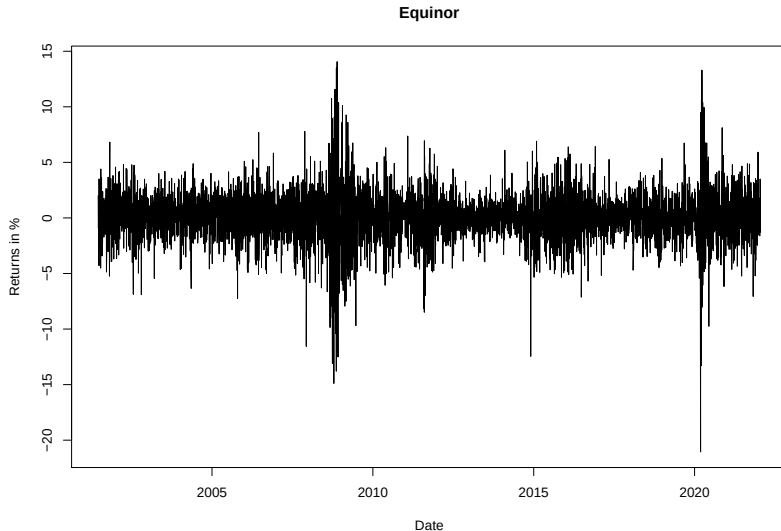
1. General information
2. Teaching model
3. Assessment
4. R as programming language
5. Material
6. Course culture

Course responsible

- ▶ Lecturer
 - ▶ Darya Yuferova, Associate Professor in Finance
 - ▶ Maximilian Rohrer, Associate Professor in Finance (Videos)
- ▶ Teaching assistant
 - ▶ Katsiaryna Falkovich, PhD student in Finance

My interest in Econometrics

...started from modelling volatility of the stock returns
(ARCH/GARCH models)



Nobel Prize in Economics 2003

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2003 was divided equally between Robert F. Engle III "for methods of analyzing economic time series with time-varying volatility (ARCH)" and Clive W.J. Granger "for methods of analyzing economic time series with common trends (cointegration)." - www.nobelprize.org



Robert F. Engle III (left) and Clive W. J. Granger at the Prize Award Ceremony at the Stockholm Concert Hall, 10 December 2003. Copyright © The Nobel Foundation 2003

Photo: Hans Mehlin

What is the course about? (i)

This course is about using data to measure a causal effect in a Financial context.

- ▶ Ideally, we would like an experiment.
- ▶ But almost always we only have observational (non-experimental) data:
 - ▶ returns to education;
 - ▶ stock prices;
 - ▶ monetary policy.

What is the course about? (ii)

- ▶ Most of the course deals with difficulties arising from using observational data to estimate causal effects
 - ▶ confounding effects (omitted factors)
 - ▶ simultaneous causality
 - ▶ “correlation does not imply causation”
 - ▶ Examples: ▶ Spurious correlation

In this course you will...

- ▶ Learn methods for estimating causal effects using observational data.
- ▶ Learn a broad set of empirical tools used in Finance.
- ▶ Focus on applications:
 - ▶ hands-on experience (a lot of coding!)
 - ▶ theory used only as needed to understand the whys of the method
- ▶ Learn to evaluate analysis by others – this means you will be able to read/understand empirical financial papers in other courses.

Financial Econometrics

- ▶ Theory is a standard econometrics theory.
- ▶ Where does Finance come in?

Financial Econometrics

- ▶ Theory is a standard Econometrics theory.
- ▶ Where does Finance come in?
 - ▶ we are going to discuss methods widely used in Finance.
 - ▶ lab sessions, academic papers used, assignments, and exam are purely Finance focused.
- ▶ No prior knowledge on Financial models is required.

Research questions

- ▶ Which stock characteristics yield a high return?
- ▶ Do prices reflect fundamental value?
- ▶ What defines the debt ratio?
- ▶ Did Elon Musk move Tesla's stock price?
- ▶ What's the impact on firm performance, if children inherit the business of their parents?
- ▶ Are people with finance related knowledge more likely to participate in the stock market?
- ▶ Does firm value increase in managerial ownership?
- ▶ etc.

Teaching model

- ▶ This course focuses on practice. All theoretical concepts are applied by implementing the actual code in R.
- ▶ Learning concepts through implementation:
 - ▶ Assignments may cover topics not yet discussed in class.
 - ▶ Understand econometrical models through coding.
- ▶ There is no correct solution which you should deduct from the set-up.
- ▶ Give you skills to make informed choices (i.e., to “torture” the data properly).

Teaching set-up

- ▶ On Tuesday at 12:00, I will publish
 - ▶ Pre-recorded video lecture(s) for the next week
 - ▶ Coding exercise without solution for the next week
- ▶ **You have to watch the video before class**
- ▶ You do not need to work on the coding exercise before class
- ▶ On Tuesdays, we have a 3 hour class (not recorded) with
 - ▶ 1h of Q&A on lecture material (reverse class-room);
 - ▶ 2h of lab session;
 - ▶ in the end of the class, I will go through the coding solution;
 - ▶ **pdf-solution will be published after the class and will be available till the end of the course.**

Assesment

- ▶ Course approval:
 - ▶ 3 assignments (Pass/Fail)
- ▶ Exam:
 - ▶ Written report (60%)
 - ▶ Oral exam (40%)

Assesment: Assignments (i)

- ▶ Three assignments.
- ▶ Groups of **four** students:
 - ▶ You can form groups on Canvas.
 - ▶ **It is your responsibility to find a group on time.**
 - ▶ Use Discussion board on Canvas to search for a group.
 - ▶ Deadline for group formation: February 10, 2026.
- ▶ Grading: Pass/Fail.
- ▶ Course approval: three “Pass” grades.
- ▶ Each assignment consists of two parts:
 - ▶ Implement analysis (1 week)
 - ▶ Peer review (1 week)

Assesment: Assignments (ii)

Implement the analysis (submit two files):

- ▶ Report (file with extension .pdf)
 - ▶ At max four pages including tables and excluding title page.
 - ▶ Interpretation of the results is important!
 - ▶ Make proper tables.
 - ▶ No copy/paste directly from R / RStudio.
- ▶ Code (file with extension .R)
 - ▶ Comment the code properly.
 - ▶ **Code without comments will automatically result in the “Fail” grade!**
- ▶ After the assignment suggested coding solution will be posted on Canvas.
- ▶ TA is responsible for grading and feedback.

Assesment: Assignmnets (iii)

- ▶ Peer review: comment on both the code and the interpretation of the assignment.
- ▶ Dual purpose:
 - ▶ help your fellow students by highlighting the most important area of improvement.
 - ▶ learn coding by reading other people's code.

Assesment: Assignmnets (iv)

- ▶ Guidelines will be provided on what should be included in peer review.
- ▶ After Assignment 1 you will receive a warning if the quality of peer review is unsatisfactory.
- ▶ Repeated insufficient quality in future peer reviews will result in a “Fail” grade for future Assignments, regardless of the quality of your own analysis.
- ▶ Peer review is an integral part of the assignment’s “Pass” or “Fail” grade.
- ▶ **The group that does not submit peer review will automatically receive a “Fail” grade.**
- ▶ Do not be judgmental.

Assesment: Assignment Schedule

- ▶ Assignment 1

- ▶ Implementation of the analysis: 24.02.2026
- ▶ Peer review: 03.03.2026

- ▶ Assignment 2

- ▶ Implementation of the analysis: 10.03.2026
- ▶ Peer review: 17.03.2026

- ▶ Assignment 3

- ▶ Implementation of the analysis: 24.03.2026
- ▶ Peer review: 31.03.2026

Final Exam (i)

- ▶ Groups of **four** students.
- ▶ Format: you will be given data and a research question.
- ▶ Mimics a real life situation a data analyst might face.
- ▶ Written report (60% of the final grade)
 - ▶ Hand-out 09:00 on 11.05.2026, submission 14:00 on 13.05.2026.
 - ▶ Take-home exam.
 - ▶ Tight formal requirements on word and table count.
 - ▶ **Group-based grade.**
 - ▶ **Grade can be repealed.**

Final Exam (ii)

Oral exam (40% of the final grade)

- ▶ Presentations will take place: 02.06.2026 – 04.06.2026
- ▶ Discuss your work and link it to econometric theory (approx. 5 min).
- ▶ Q&A on theory presented in class (approx. 20 min).
- ▶ **Grades are individual.**
- ▶ **Grades cannot be repealed.**
- ▶ Objectivity ensured through the presence of TA who will take notes of your performance.
- ▶ Feedback in the week after the oral exam.

Final Exam (iii)

The grading is based on

- ▶ the correctness of the results.
- ▶ the clarity and persuasiveness of the argument.
- ▶ the student's ability to learn and implement new concepts.

R as a programming language

- ▶ R is an open source and widely used programming language.
- ▶ One of the most popular languages for data analysis.
- ▶ Not only used for statistics but very flexible language:
 - ▶ textual data analysis
 - ▶ machine learning
 - ▶ pattern recognition
 - ▶ etc.
- ▶ Programming skills are transferable
 - ▶ Python (open source)
 - ▶ Matlab (commercial)
- ▶ **There is no unique coding solution.**

Material

- ▶ Videos / Slides / Code will be available on Canvas.
- ▶ Textbooks:
 - ▶ Stock and Watson, Introduction to Econometrics, Global Edition, 4th edition
 - ▶ Heiss, Using R for Introductory Econometrics
- ▶ Software: R and RStudio

Course culture (i)

- ▶ Questions regarding course organization, content, and assignments should be posted at Discussion board on Canvas under respective threads.
- ▶ We (my TA and I) will **NOT** reply to questions regarding course organization, content, and assignments via e-mail!
- ▶ TA has no authority to answer questions other than explaining course content / coding.

Course culture (ii)

- ▶ Learning coding is about trial and error, in particular finding mistakes within code (“debugging”).
- ▶ We will help you with your code during lab sessions.
- ▶ However, outside of class we will not correct your code, especially not per e-mail.

Canvas

- ▶ In general, all communication takes place via Canvas.
- ▶ Detailed course schedule is available on Canvas.
- ▶ Group formation: please sign-up on Canvas/People.

R and RStudio

- ▶ Please install R and RStudio:
 - ▶ R: <https://cran.r-project.org/>
 - ▶ RStudio: <https://rstudio.com/products/rstudio/download/>
- ▶ Please install packages we are going to use in this course (list of packages available on Canvas).

The rest of today

- ▶ Katsiaryna (my TA) will help you with:
 - ▶ R/RStudio installation;
 - ▶ packages installation.

Next class

- ▶ Introduction to R by Katsiaryna.
- ▶ For those of you who know R and R-Studio, have a look at the lab session exercises.
- ▶ If you can solve those tasks easily, you do not need to watch the recorded lecture and attend the next session.